DP Biology – B3.3 (HL only)

Muscle and motility

- How do muscles contract and produce movement?
- What do animals gain from having muscle tissue?

HL statements B3.3.1–10. Textbook pp. 326–340.

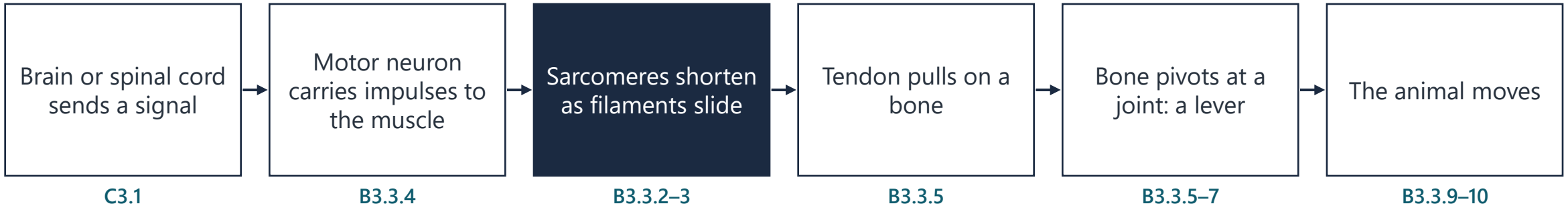


Image: a humpback whale breaching. Giles Laurent, CC BY-SA 4.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B3.3.1, B3.3.9	Compare motile and sessile organisms, and explain four reasons for locomotion with an example of each
2	B3.3.2	Describe the structure of a muscle fibre and a sarcomere, and identify the bands in a micrograph
3	B3.3.2	Explain how a nerve impulse leads to contraction, using the cross-bridge cycle and the roles of Ca^{2+} and ATP
4	B3.3.3, B3.3.8	Explain how titin and antagonistic muscles return a muscle to its length, using the intercostal muscles
5	B3.3.4	Describe a motor unit and explain how the nervous system controls the force of a contraction
6	B3.3.5–7	Explain skeletons as anchors and levers, movement at the hip joint, and how range of motion is measured
7	B3.3.10	Explain how marine mammals are adapted for swimming and for breathing between dives

The big picture: from molecule to movement



Every movement an animal makes depends on three systems working together. The **nervous system** decides when and how strongly to move. **Muscles** turn chemical energy from ATP into a pulling force. The **skeleton** gives muscles something to pull against, and its joints and levers turn a short pull into a useful movement.

A muscle can only pull, never push. That one fact explains why muscles come in pairs, why titin matters, and how joints are arranged.

Big question for this chapter

How does a reaction in a protein a few nanometres long become a movement big enough to lift your arm, or drive a whale through the water?

Movement is universal, even for organisms that stay put

B3.3.1



Barnacles: sessile adults that sweep food in with their legs. Chris Light, CC BY-SA 4.0.

Motile organisms move from place to place (locomotion). **Sessile** organisms are fixed to one spot, but they still move parts of their bodies.

A **barnacle** glues its head to a rock and spends its adult life there. It feeds by sweeping feathery legs (cirri) through the water and pulling them back to its mouth. Corals catch prey with tentacles armed with stinging cells.

Many sessile animals have a **motile larva** that swims away and settles somewhere new, so the species can still spread.

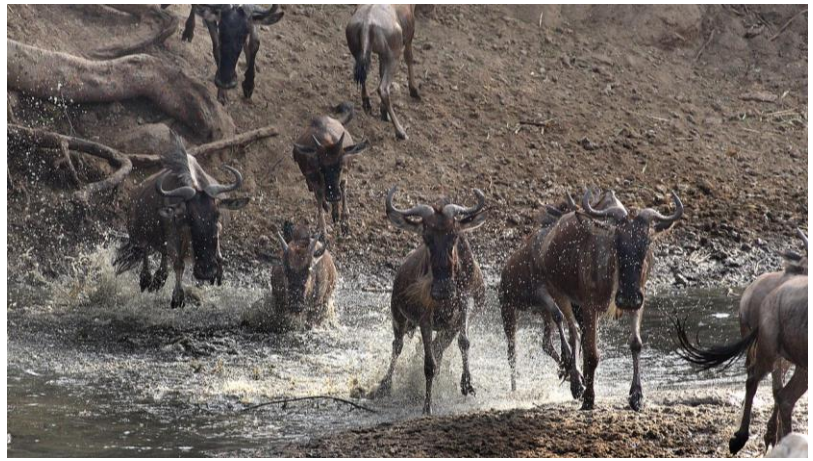
	Motile	Sessile
Example	Cheetah, whale, most insects	Barnacle, coral, sponge
Adaptations	Limbs, fins or wings; streamlining; large muscle mass	Attachment to a surface; protective shell; feeding structures that move

Why animals move from place to place

B3.3.9



Cheetah chasing prey. Hein waschefort, CC BY-SA 3.0.

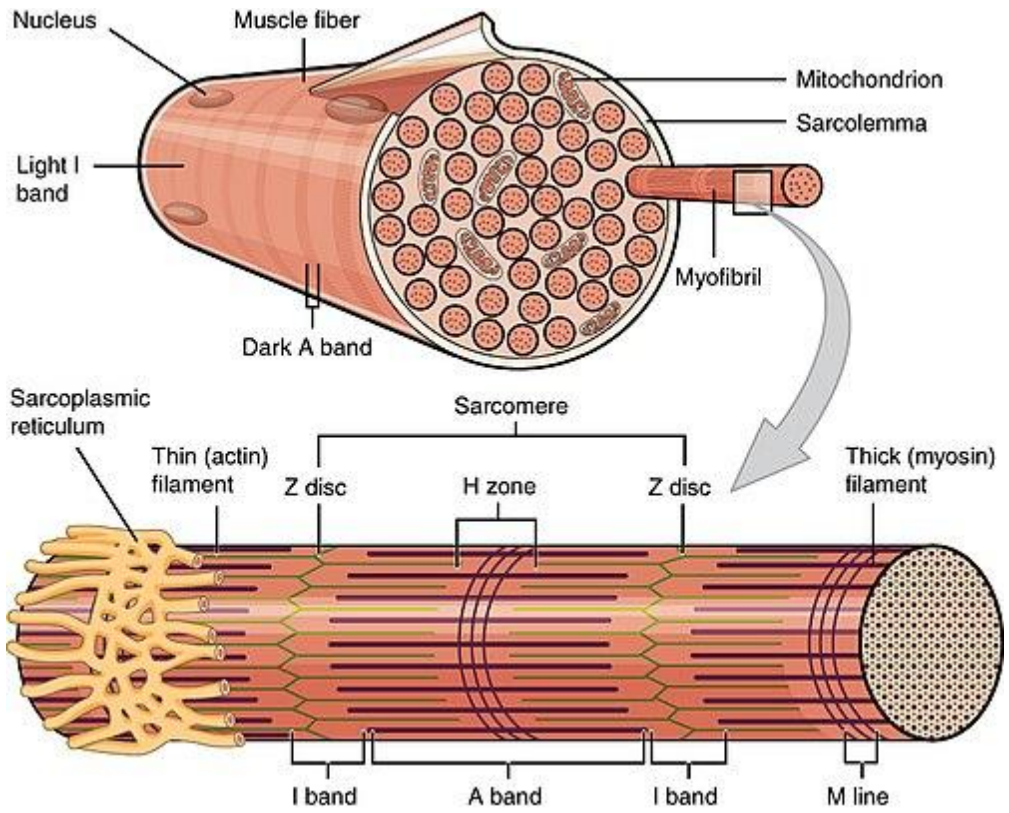


Wildebeest migrating in the Serengeti. Wolves201, CC BY-SA 4.0.

Reason	Example and what it achieves
Foraging for food	A **cheetah** sprints at up to about 100 km h^{-1} to catch gazelles; a blackbird hops across a lawn to find earthworms
Escaping danger	A **snowshoe hare** zig-zags away from a lynx; a squid jets backwards by squeezing water out of its mantle
Searching for a mate	Solitary **orangutans** travel long distances through the forest to meet a partner; male moths fly upwind towards a female's scent
Migration	**Wildebeest** follow the rains to fresh grass; **humpback whales** swim thousands of km between cold feeding waters and warm breeding waters

From whole muscle down to the filaments

B3.3.2



A skeletal muscle fibre. OpenStax, CC BY 4.0.

Level	What it is
Muscle	Bundles of fibres wrapped in connective tissue; tendons at each end
Muscle fibre	One long cell, 10–100 µm wide and often several cm long, with **many nuclei**
Myofibril	Cylinders about 1–2 µm wide, packed side by side inside the fibre
Sarcomere	The repeating unit of a myofibril, about 2–2.5 µm long at rest
Filaments	Thick **myosin** (about 15 nm wide) and thin **actin** (about 7 nm wide)

Why it matters

Thousands of sarcomeres sit end to end in each myofibril. If each one shortens a little, the whole fibre shortens a lot.

A muscle fibre is a cell built for contraction

B3.3.2

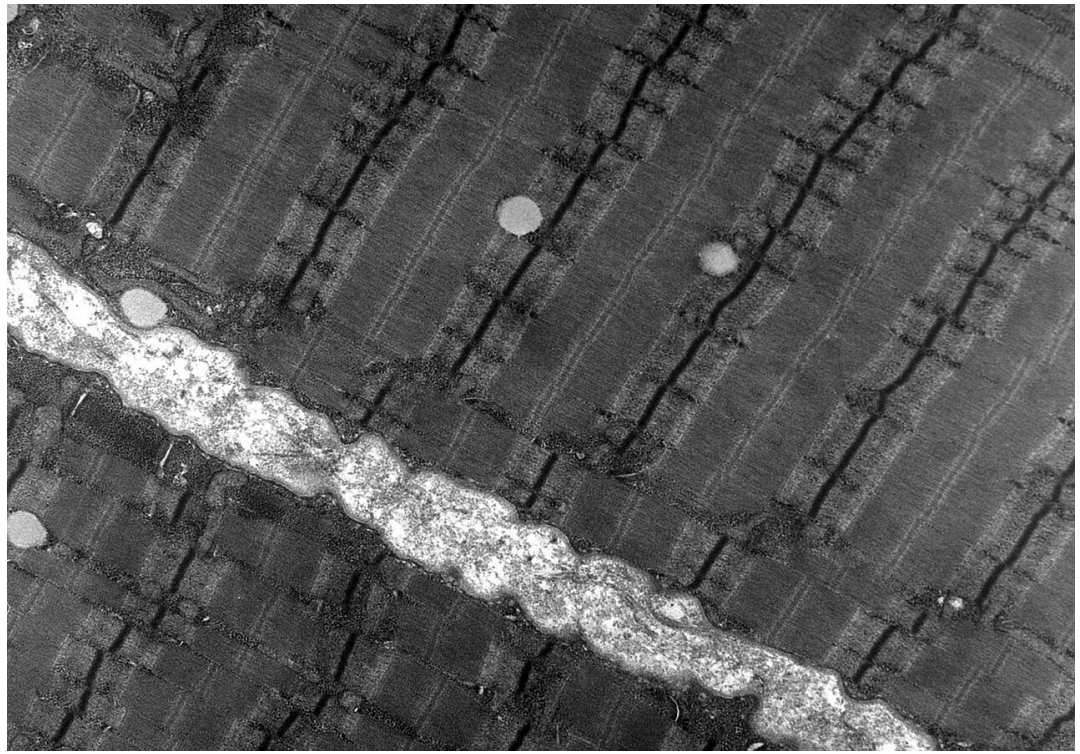
Feature	What it is	Why it helps
Sarcolemma	The plasma membrane of the fibre	Carries action potentials over the whole surface
T-tubules	Folds of the sarcolemma that run deep into the fibre	Carry the impulse inside, so every myofibril is triggered at once
Sarcoplasmic reticulum (SR)	A network of specialised smooth ER wrapped around each myofibril	Stores Ca^{2+} and releases it when the fibre is stimulated
Many mitochondria	Packed between the myofibrils	Supply ATP by aerobic respiration
Many nuclei	Spread along the edge of the fibre	Control protein synthesis along a very long cell

Common mistake

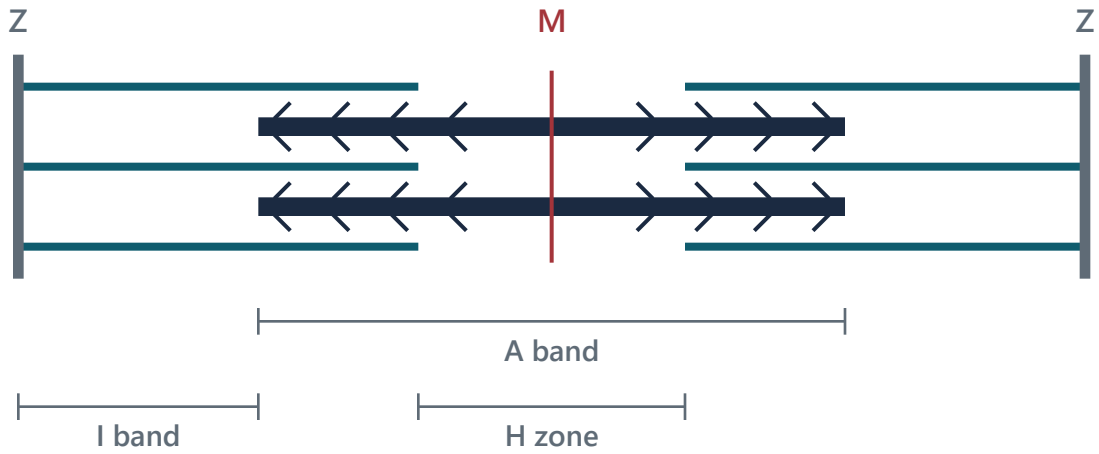
Actin and myosin are made by free ribosomes in the cytoplasm, not in the sarcoplasmic reticulum, which has no ribosomes.

The sarcomere: why skeletal muscle is striped

B3.3.2



Electron micrograph of human skeletal muscle: repeating sarcomeres. Louisa Howard, public domain.



Relaxed sarcomere (drawn to scale)

A sarcomere runs from one **Z line** to the next. Thin **actin** filaments are anchored to the Z lines; thick **myosin** filaments sit in the middle, held at the **M line**.

The dark **A band** is the length of the myosin. The light **I band** has actin only. The **H zone** has myosin only.

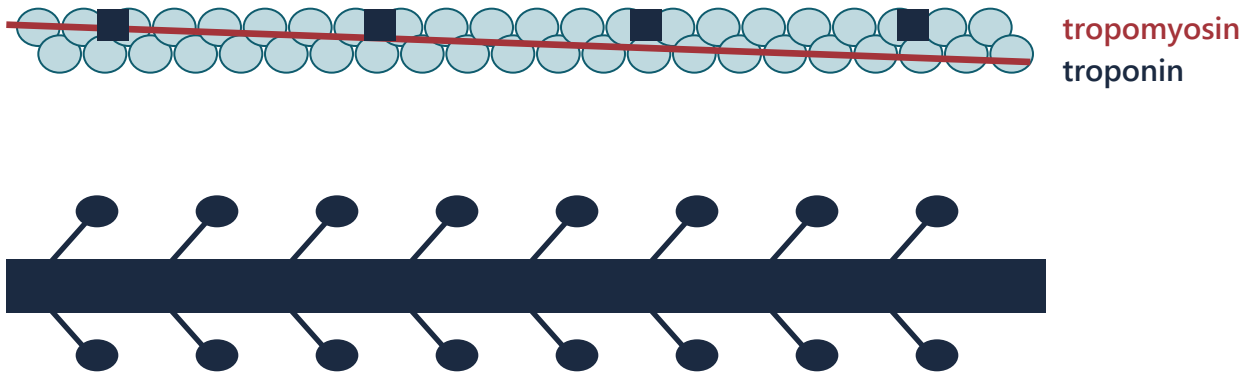
Try it

On the micrograph, find two Z lines and one A band. Estimate how many sarcomeres there are across the image.

Actin, myosin and the proteins that control them

B3.3.2

Thin filament: actin, with tropomyosin and troponin



Thick filament: myosin tails bundled, heads sticking out

Actin is a chain of globular units twisted into a double strand. Each unit has a **binding site** for a myosin head.

Tropomyosin is a thread that lies along the actin and covers those binding sites when the muscle is at rest.

Troponin is attached to tropomyosin. When Ca^{2+} binds to it, it pulls tropomyosin aside and the sites are exposed.

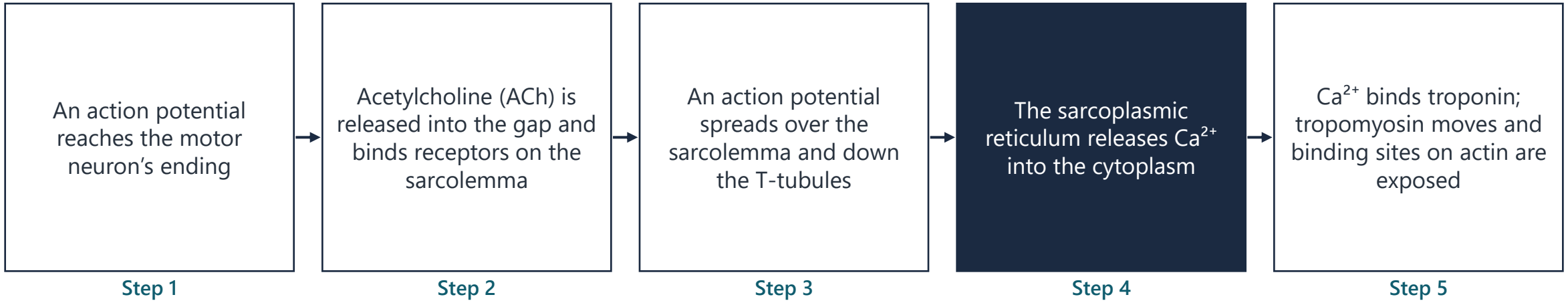
Myosin molecules have long tails and a head that binds actin and hydrolyses ATP (it is an ATPase).

Key idea

Tropomyosin is the lock and calcium is the key. At rest, myosin heads are ready to bind but cannot reach the sites on actin. A rise in Ca^{2+} inside the fibre moves the blocker, and contraction can start at once. A fall in Ca^{2+} blocks the sites again and the muscle relaxes.

From nerve impulse to calcium release

B3.3.2



The junction between a motor neuron and a muscle fibre is the **neuromuscular junction**. It works like any chemical synapse: the impulse is passed on by a neurotransmitter, here acetylcholine.

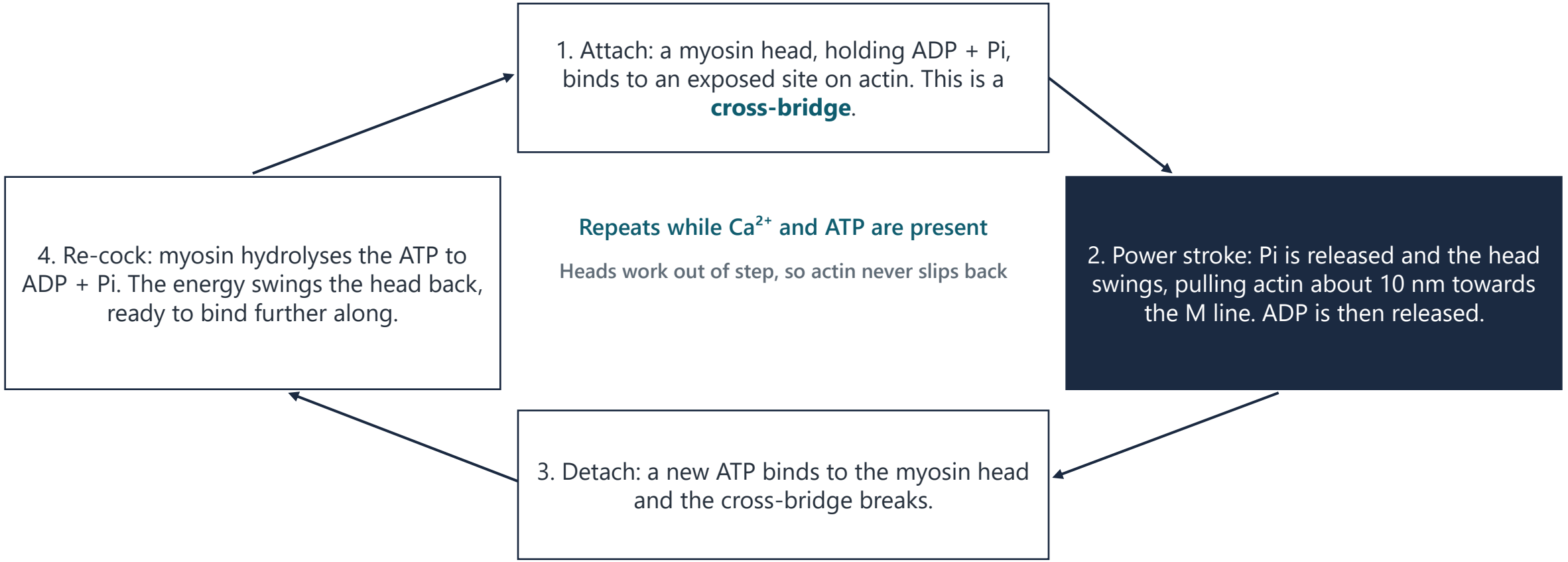
The whole sequence takes a few milliseconds. Because the T-tubules carry the signal deep into the fibre, every sarcomere in the fibre is activated at almost the same moment.

Link

C2.2 Neural signalling: release of neurotransmitter by exocytosis, and how the impulse is stopped. Acetylcholinesterase breaks ACh down, so one impulse gives one brief contraction.

The cross-bridge cycle: how myosin pulls actin

B3.3.2

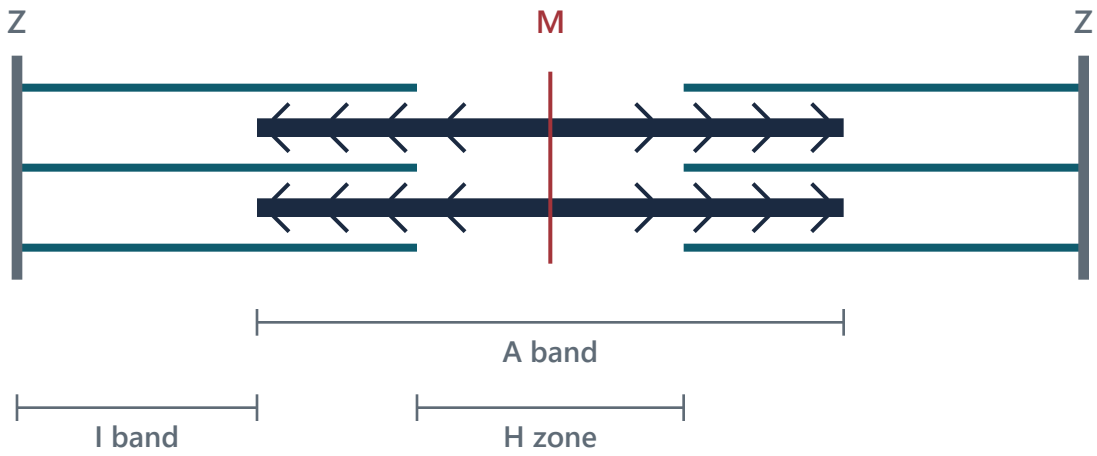


Two jobs for ATP

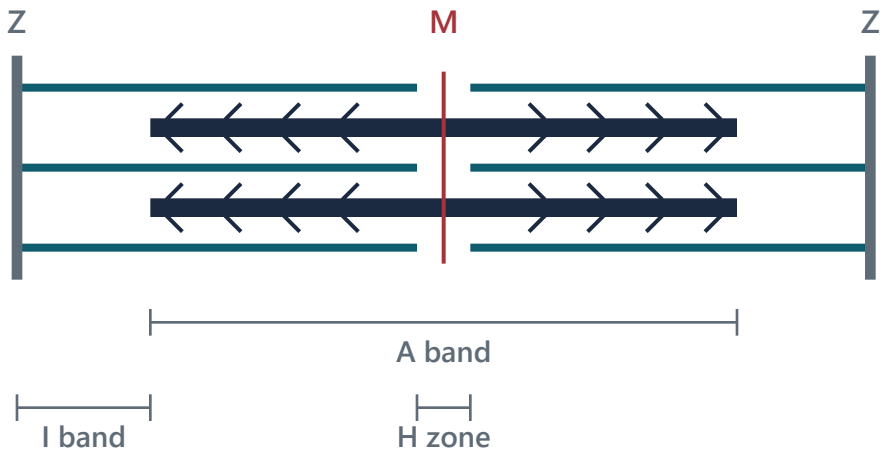
ATP **breaks** the cross-bridge (step 3) and its hydrolysis **re-cocks** the head (step 4). Each head uses one ATP per cycle, and each head cycles several times a second.

What gets shorter when a sarcomere contracts?

B3.3.2



Relaxed



Contracted

Part	In contraction	Why
Sarcomere, I band, H zone	Shorter	Actin slides in over the myosin, so the Z lines move closer
A band, actin and myosin	No change	The filaments slide; they do not shorten themselves

Relaxation, and what happens without ATP

B3.3.2

When impulses stop arriving, acetylcholine is broken down and the sarcolemma is no longer stimulated.

Ca^{2+} pumps in the SR membrane use ATP to move calcium back into the SR. As Ca^{2+} falls, troponin lets tropomyosin cover the binding sites again.

No new cross-bridges can form, so the fibre stops pulling. It does not lengthen by itself: it is stretched back by an **antagonistic muscle**, helped by the recoil of **titin**.

Rigor mortis

After death, respiration stops and ATP runs out. Myosin heads stay attached to actin because there is no ATP to break the cross-bridges, so muscles become stiff a few hours after death. This is strong evidence that ATP is needed to **detach** myosin, not only to power the stroke.

Cramp

If Ca^{2+} is not removed quickly enough, for example in tired muscle, cross-bridges keep forming and the muscle stays contracted.

Remember

ATP is needed for contraction (power stroke and detachment) and for relaxation (pumping Ca^{2+} back).

Check-in: muscle structure and contraction

B3.3.2

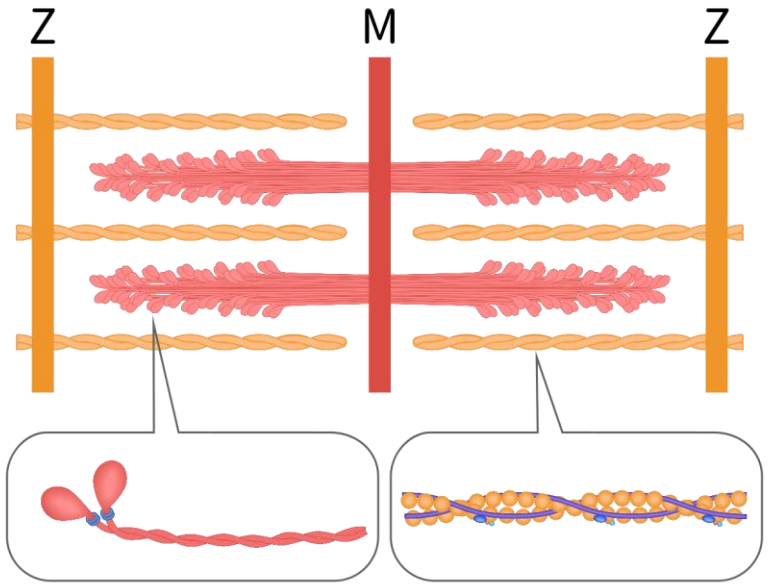
1. Name the store of calcium ions inside a muscle fibre.
2. State the role of troponin.
3. A micrograph shows sarcomeres 1.8 μm long with no visible H zone. Is the muscle relaxed or contracted? Give a reason.
4. Explain why a muscle fibre needs ATP even to relax.
5. Outline the events that follow binding of ATP to a myosin head.

How to use this

Answer on mini whiteboards or in your notes, without looking back. Then compare with a partner.

Titin: a molecular spring inside the sarcomere

B3.3.3



Titin is the largest protein known. One molecule has over 30 000 amino acids and stretches from the Z line to the M line, about half a sarcomere.

Part of titin is folded into a coil that behaves like a **spring**. When the sarcomere is stretched by another muscle, titin is extended and stores potential energy. When the stretch stops, titin **recoils** and helps the sarcomere return to its resting length.

Titin also holds each myosin filament in the centre of the sarcomere and stops the sarcomere being **overstretched**, which would pull the filaments apart.

Sarcomere with titin (pink springs) and actin. DBCLS, CC BY 4.0.

Link to the next slides

Titin explains how a stretched muscle springs back, but something must stretch it first. That is the job of the antagonistic muscle.

Antagonistic muscles: why muscles work in pairs

B3.3.3

A muscle can only exert force by contracting, which **pulls**. It cannot push itself back to its original length.

So muscles are arranged in **antagonistic pairs** across a joint. When one muscle (the agonist) contracts, the other (the antagonist) relaxes and is stretched. To reverse the movement, the roles swap.

The stretching of the relaxed muscle is what stores energy in its titin, ready for its own recoil.

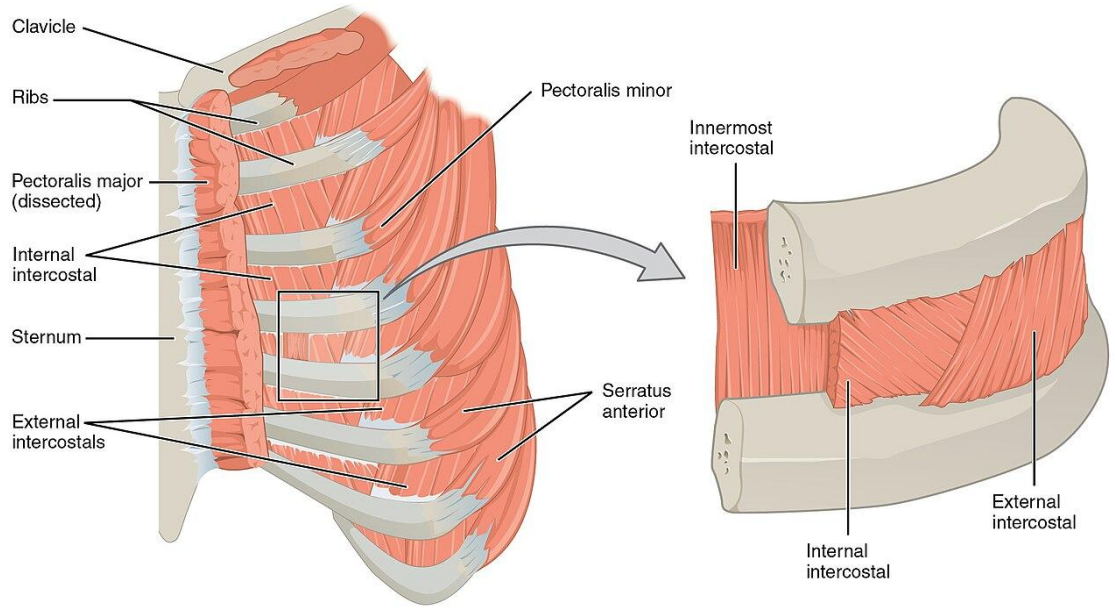
Joint	Flexor (bends)	Extensor (straightens)
Human elbow	Biceps	Triceps
Human knee	Hamstrings	Quadriceps
Insect leg joint	Flexor muscle inside the exoskeleton	Extensor muscle inside the exoskeleton

Think about it

Both muscles of a pair often contract a little at the same time. Suggest why this is useful when you hold a full cup of tea steady, or when a gymnast holds a position.

Intercostal muscles: an antagonistic pair inside the body

B3.3.8



Internal and external intercostal muscles. OpenStax, CC BY 4.0.

Between each pair of ribs are two layers of muscle whose fibres run in **different directions**, roughly at right angles to each other.

The **external** intercostals pull the ribs **up and out**. The **internal** intercostals pull them **down and in**.

Because the fibres slant in opposite directions, contracting one layer stretches the other, storing energy in its titin.

Why this matters

This is an example of antagonistic muscles moving a part of the body, not the whole animal. The same principle that bends your elbow also changes the volume of your thorax, and so the pressure in your lungs: a bigger thorax means lower pressure, so air flows in; a smaller thorax means higher pressure, so air is pushed out.

Which muscles work in quiet and in forced breathing?

B3.3.8

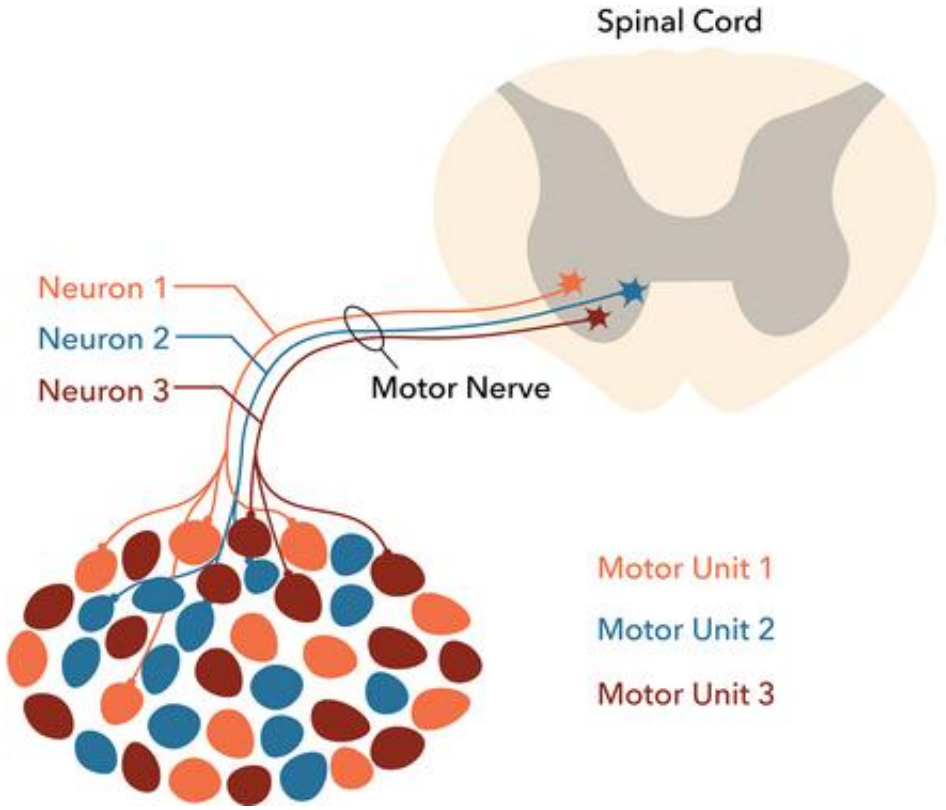
Phase	Muscles that contract	Effect on the ribs
Quiet breathing in	Diaphragm (main muscle); external intercostals help	Diaphragm flattens; ribs rise a little
Quiet breathing out	None: the diaphragm and external intercostals relax and the stretched lungs recoil	Ribs fall back passively
Forced breathing in	Diaphragm and external intercostals strongly, helped by neck muscles	Ribs pulled well up and out
Forced breathing out	**Internal intercostals** and abdominal muscles	Ribs pulled down and in; diaphragm pushed up

Key idea

The internal intercostals are mainly used when breathing out hard, for example during exercise or when coughing.

Motor units: one neuron and the fibres it controls

B3.3.4



Three motor units in one muscle. D. Walsh and A. Sved, CC BY-SA 4.0.

A **motor unit** is one motor neuron together with all the muscle fibres it stimulates.

The motor neuron's cell body is in the spinal cord or brain stem. Its axon, insulated by a **myelin sheath**, branches near the muscle, and each branch ends at a **neuromuscular junction** on one fibre.

Each fibre is supplied by only one neuron. When that neuron fires, **every fibre in the unit contracts together**.

The fibres of one unit are scattered through the muscle, so it pulls evenly.

Notice

In the diagram, fibres of the three units are mixed together. Firing one neuron gives a weak, even pull; firing all three gives a strong one.

Controlling the strength of a contraction

B3.3.4

	Small motor units	Large motor units
Fibres per neuron	Around 10	Hundreds to about 2000
Example muscles	Muscles that move the eye; muscles of the fingers	Thigh and calf muscles; back muscles
Type of control	Fine, precise movements	Powerful but less precise movements

A single motor unit contracts fully or not at all. A muscle produces more force by **recruiting** more motor units: small units are used first, and large units are added as more force is needed. Firing impulses faster also lets twitches add together into a stronger, steady pull.

Even at rest a few units are always active, giving muscles their **tone** and keeping posture.

Check-in: titin, antagonists, intercostals and motor units

B3.3.3–4

1. State two roles of titin.
2. Explain why a muscle cannot return to its resting length on its own.
3. Describe how the internal and external intercostal muscles act as an antagonistic pair.
4. Define a motor unit.
5. Suggest why the muscles that move the eye have very small motor units.

How to use this

Write full sentences. Swap with a partner and mark each other's answers using the notes slides.

Skeletons give muscles something to pull against

B3.3.5

	Exoskeleton	Endoskeleton
Found in	Arthropods: insects, spiders, crustaceans	Vertebrates: fish, amphibians, reptiles, birds, mammals
Made of	**Chitin**, often hardened with protein (and calcium carbonate in crustaceans)	**Bone** (or cartilage in sharks and rays)
Muscles attach	To the **inside** of hollow tubes, across flexible joints	To the **outside** of bones, by tendons
Growth	Must be shed (moulting) for the animal to grow	Grows with the animal

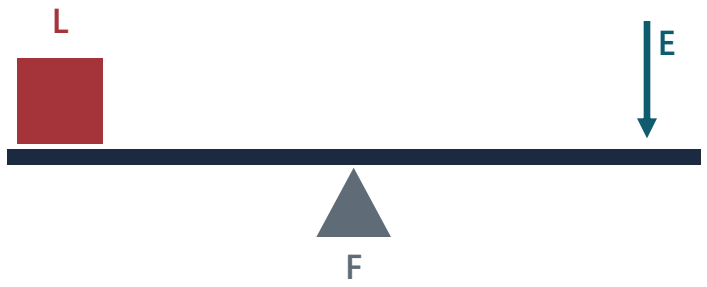
Both types do the same jobs

Both support the body, protect organs and provide **anchorage** for muscles. In both, muscles work in antagonistic pairs across joints, and the rigid parts act as levers.

Bones as levers

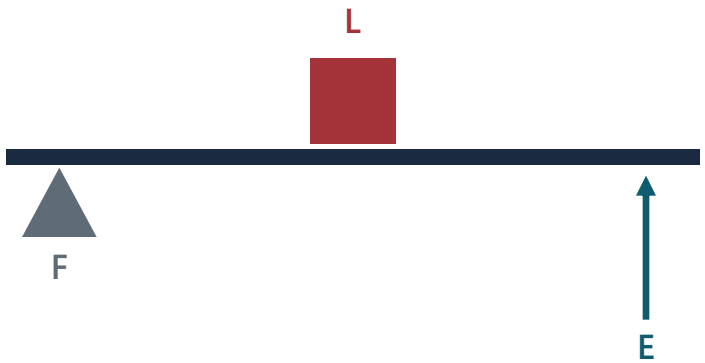
B3.3.5

First class



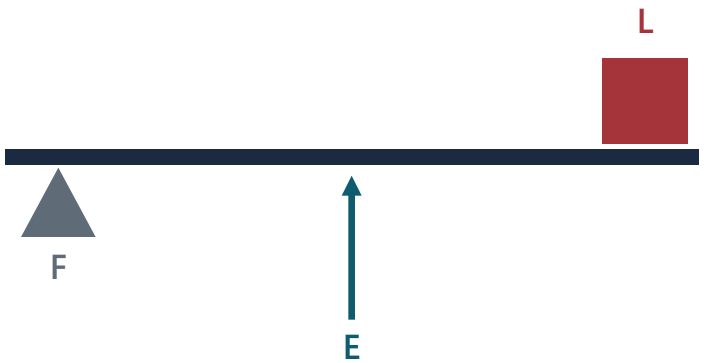
F in the middle: nodding the head on the top of the spine

Second class



L in the middle: rising onto the toes

Third class



E in the middle: the biceps lifting the forearm

In a lever, the joint is the pivot or **fulcrum** (F), the muscle supplies the **effort** (E) where its tendon attaches, and the **load** (L) is the weight being moved.

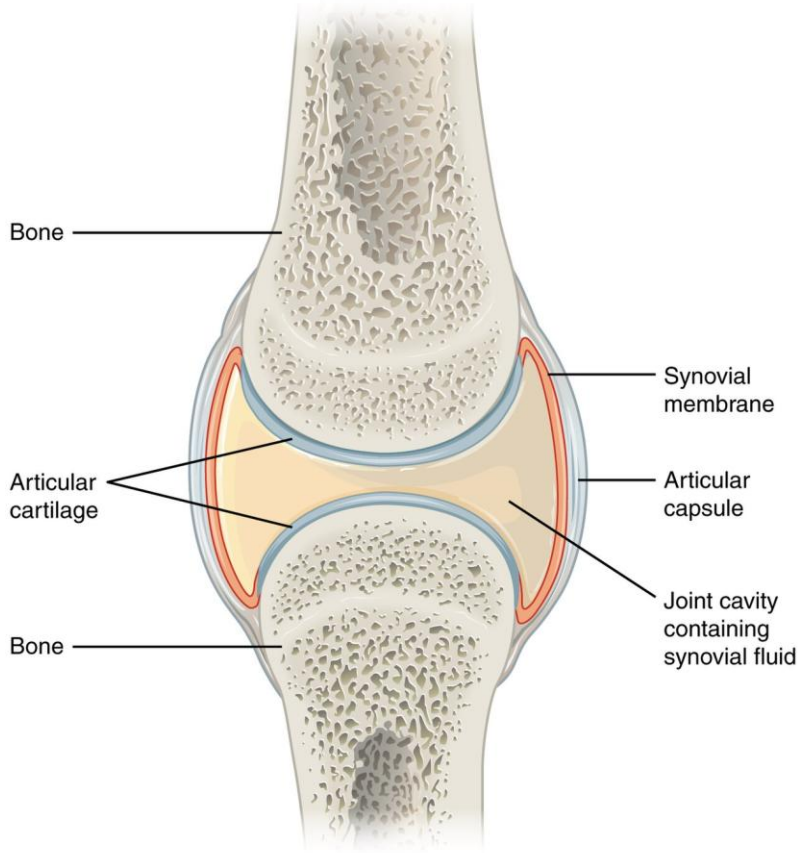
Most limbs are **third-class** levers. The biceps attaches about 4 cm from the elbow, but the hand is about 30 cm away. The muscle must pull about 7 times harder than the load, but the hand moves 7 times further and faster.

Trade-off

Muscles attached close to a joint give speed and range of movement, at the cost of needing a large force. That is why limb muscles are so strong.

Synovial joints: built to move with little friction

B3.3.6

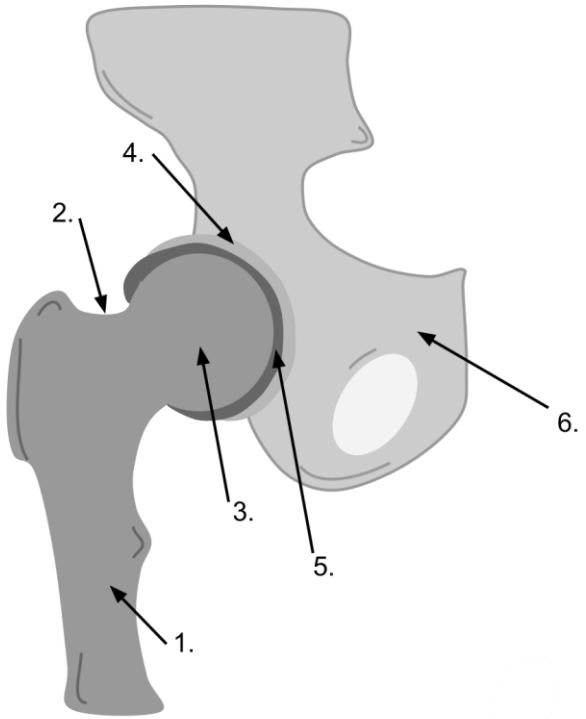


Structure of a synovial joint. OpenStax, CC BY 3.0.

Part	Role
Bones	Rigid levers that move relative to each other
Cartilage	Smooth, slightly springy layer on the bone ends: reduces friction and absorbs shock
Synovial fluid	Lubricates the joint and feeds the cartilage, which has no blood supply
Joint capsule and synovial membrane	Encloses the joint; the membrane secretes the fluid
Ligaments	Tough fibres joining **bone to bone**: hold the joint together and limit movement
Muscles and tendons	Muscles pull; tendons join **muscle to bone**

The hip: a ball-and-socket joint

B3.3.6



Hip joint: head of the femur in the socket of the pelvis. Kcotton15, CC BY-SA 4.0.

At the hip, the rounded **head of the femur** (thigh bone) fits into a deep cup in the **pelvis**. Both surfaces are covered with cartilage and bathed in synovial fluid.

Muscles crossing the joint, attached by tendons, move the leg in **three planes**: forwards and back, out to the side and in, and rotation.

The socket is deep and the ligaments are very strong, so the hip is stable enough to carry the body's weight. The shoulder is also a ball-and-socket joint, but its socket is shallow: it moves more freely but dislocates more easily.

Exam tip

You must be able to name the femur and the pelvis. You do not need to name the muscles or ligaments of the hip.

Range of motion depends on the type of joint

B3.3.7



A goniometer. Rama, CC BY-SA 2.0 fr.

Joint	Type	Movements	Typical range
Hip	Ball and socket	Three planes, with rotation	Flexion about 120°
Shoulder	Ball and socket	Three planes; widest range	Flexion about 180°
Knee	Hinge	Mainly one plane: flexion and extension	Flexion about 135°
Elbow	Hinge	One plane: flexion and extension	Flexion about 150°

The shape of the bone surfaces and the position of the ligaments decide which movements are possible. Range of motion is measured with a **goniometer**: two arms joined at a hinge with a protractor, lined up along the two bones with the hinge over the joint. Phone apps use the phone's motion sensors in the same way.

Variables to control

Starting position, which movement is measured, warm-up, and the person (age, sex, flexibility and injury all affect range).

Working with data: knee flexion

B3.3.7

Students measured the knee flexion of five volunteers three times each with a goniometer. These data are invented for practice; the values are realistic.

Volunteer	Trial 1 / °	Trial 2 / °	Trial 3 / °
A	128	131	129
B	135	134	136
C	122	140	123
D	130	129	131
E	126	127	125

1. Calculate the mean for volunteer A.
2. Identify the anomalous result, and explain how you would deal with it.
3. Suggest one reason the trials for the same person differ.
4. Suggest how to make the method more reliable.

Swimming mammals: cutting through water

B3.3.10



Common bottlenose dolphin swimming. CC0.

Water is about 800 times denser than air, so drag is the main problem for a swimming animal. Whales, dolphins, seals and sea lions are all descended from land mammals and have become adapted in similar ways.

A **streamlined** (torpedo-shaped) body lets water flow smoothly past. There are few things that stick out: no external ears in whales and dolphins, internal reproductive organs, a short and stiff neck, and almost no hair.

A thick layer of **blubber** smooths the body outline and insulates. The skin sheds cells and secretes oil, keeping its surface smooth.

Link

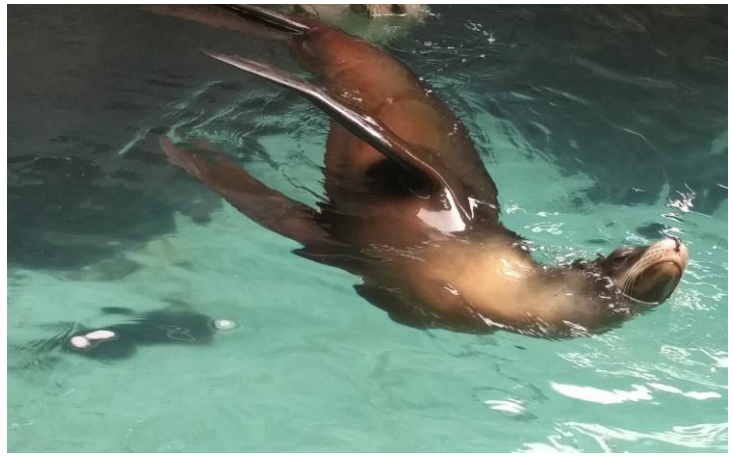
A1.1 Water: water's high density, viscosity and thermal conductivity are all challenges for aquatic mammals. Blubber helps with buoyancy as well as heat loss.

Flippers, flukes and breathing between dives

B3.3.10



Blowhole of a bottlenose dolphin. Public domain.



California sea lion. ThePenultimateOne, CC BY-SA 4.0.

Adaptation	How it helps
Flippers	Forelimbs with the same bones as a human arm, but short and paddle-shaped: used for steering and balance (sea lions also use them to swim)
Fluke	A boneless tail fin, flat and **horizontal**, beaten **up and down** by powerful back muscles. Fish tails are vertical and move side to side
Hind limbs	Lost in whales and dolphins (only a small pelvic bone remains); seals keep hind flippers and sweep them side to side
Airway	Nostrils moved to the top of the head as a **blowhole**, closed by muscle during a dive. The airway is separate from the gut, so the animal can open its mouth under water

Common mistakes to avoid

Not this	But this
Muscles push bones back into place.	Muscles only pull. An antagonistic muscle pulls the bone back.
Actin and myosin shorten during contraction.	The filaments slide past each other. The sarcomere shortens; the filaments do not.
Calcium ions form the cross-bridges.	Ca^{2+} binds troponin, which moves tropomyosin. Myosin heads form the cross-bridges.
ATP is only needed to make the power stroke.	ATP is also needed to break cross-bridges and to pump Ca^{2+} back into the SR.
A motor unit is one neuron and one fibre.	It is one neuron and all the fibres it supplies.
Ligaments join muscle to bone.	Ligaments join bone to bone. Tendons join muscle to bone.

Check that you can...

1. **Compare** motile and sessile organisms, and **outline** four reasons for locomotion with an example of each
2. **Draw** and label a sarcomere, and **identify** the bands in an electron micrograph
3. **Explain** how a nerve impulse leads to Ca^{2+} release and contraction
4. **Describe** the cross-bridge cycle, including both roles of ATP
5. **Explain** the roles of titin and antagonistic muscles, using the intercostal muscles
6. **Describe** a motor unit and how the force of a contraction is controlled
7. **Distinguish** between exoskeletons and endoskeletons, and **explain** bones as levers
8. **Outline** the roles of the parts of the hip joint, and **compare** range of motion in different joints
9. **Explain** the adaptations of marine mammals for swimming and breathing

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Explain the role of calcium ions in muscle contraction.
2. State what happens to the A band, the I band and the H zone when a sarcomere contracts.
3. Outline two adaptations of whales for swimming.

Image credits

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